

MODULE 1: WHAT IS EXCAVATION?

मॉड्यूल 1: खुदाई (Excavation) क्या है?

English

Excavation means any man-made cut, trench, pit, cavity or depression made in the ground.

हिन्दी

खुदाई (Excavation) का अर्थ है जमीन में बनाई गई कोई भी खाई, गड्ढा, कटिंग या अवसाद।

Examples

- ✓ Foundation excavation
- ✓ Pipeline trench
- ✓ Cable trench
- ✓ Drain excavation
- ✓ Pit excavation

MODULE 2: WHY EXCAVATION IS DANGEROUS?

मॉड्यूल 2: खुदाई कार्य खतरनाक क्यों है?

Major Hazards

- ✗ Soil collapse
- ✗ Fall of persons into pit
- ✗ Damage to underground cables
- ✗ Gas leakage
- ✗ Water flooding
- ✗ Electrocution
- ✗ Equipment rollover
- ✗ Falling materials

Golden Rule

Excavation can become fatal if proper precautions are not taken.

यदि उचित सुरक्षा उपाय नहीं अपनाए गए तो खुदाई कार्य जानलेवा हो सकता है।

MODULE 3: EXCAVATION PERMIT & CLEARANCE

MODUL 3: खुदाई अनुमति और क्लियरेंस

Before excavation:

- ☒ Six Party Clearance
- ☒ Excavation Plan
- ☒ Utility Mapping
- ☒ Cable Detection
- ☒ Trial Trench
- ☒ Supervisor Approval

Rule

No Clearance = No Excavation

क्लियरेंस नहीं = खुदाई नहीं

MODULE 4: UNDERGROUND UTILITIES

मॉड्यूल 4: भूमिगत सेवाएँ

Before excavation identify:

- ☒ Electrical Cables
- ☒ FO Cables
- ☒ Telephone Cables
- ☒ Water Pipelines

- ✓ Gas Pipelines
- ✓ Sewer Lines
- ✓ Storm Water Lines

हिन्दी

खुदाई शुरू करने से पहले सभी भूमिगत सेवाओं की पहचान अनिवार्य है।

MODULE 5: TRIAL TRENCH

मॉड्यूल 5: ट्रायल ट्रेंच

Purpose

To identify underground cables and utilities.

Rules

- ✓ Trial trench must be excavated manually.
- ✓ Cable detector must be used.
- ✓ Stop immediately if any utility is found.

Never

- ✗ Direct mechanized excavation without clearance.

MODULE 6: SOIL TYPES

मॉड्यूल 6: मिट्टी के प्रकार

Type A

Strong cohesive soil.

मजबूत मिट्टी।

Type B

Previously disturbed soil.

ढीली या पहले से खुदी हुई मिट्टी।

Type C

Loose sand, wet soil, backfilled soil.

रेतीली, गीली और कमजोर मिट्टी।

Important

Type C soil is the most dangerous for cave-ins.

MODULE 7: COMMON EXCAVATION HAZARDS

मॉड्यूल 7: प्रमुख खुदाई खतरे

-  Cave-in
-  Toppling
-  Sliding
-  Boiling
-  Bulging
-  Water accumulation
-  Falling materials
-  Falling persons

These conditions can quickly lead to injury or fatality.

MODULE 8: BARRICADING REQUIREMENTS

मॉड्यूल 8: बैरिकेडिंग

Requirements

-  Barricades around excavation
-  Warning signs

✓ Red lights during night

✓ Restricted access

✓ Proper walkways

Important

Barricades must remain throughout excavation work.

MODULE 9: SPOIL PILE MANAGEMENT

मॉड्यूल 9: खुदाई सामग्री का प्रबंधन

Rule

Spoil material must be kept at least 1 metre from excavation edge.

Do's

✓ Keep materials away

✓ Prevent rollback

✓ Maintain berm

Don'ts

✗ Store materials near edge

✗ Dump soil inside excavation

MODULE 10: SOIL COLLAPSE PROTECTION

मॉड्यूल 10: मिट्टी धंसने से सुरक्षा

Three Protection Methods

✓ Sloping

✓ Benching

✓ Shoring

These methods prevent collapse of trench walls.

MODULE 11: BENCHING

मॉड्यूल 11: बेंचिंग

English

Benching creates steps in the excavation wall to improve stability.

हिन्दी

बेंचिंग में खुदाई की दीवार को सीढ़ीनुमा बनाया जाता है जिससे मिट्टी धंसने का खतरा कम होता है।

Important

Benching becomes necessary for deeper excavations.

MODULE 12: SHORING

मॉड्यूल 12: शोरिंग

English

Shoring uses support structures to prevent trench wall collapse.

हिन्दी

शोरिंग में लकड़ी, स्टील या अन्य सहारे का उपयोग कर मिट्टी धंसने से बचाया जाता है।

Mandatory

Deep trenches require competent person approval.

MODULE 13: WATER ACCUMULATION

मॉड्यूल 13: पानी भरना

Hazards

 Flooding

✗ Soil weakening

✗ Cave-in

✗ Drowning

Requirements

✓ Dewatering Pumps

✓ Drainage System

✓ Continuous Monitoring

Rule

No work inside water-filled excavation.

MODULE 14: SAFE ENTRY & ACCESS

मॉड्यूल 14: सुरक्षित प्रवेश

Requirements

✓ Ladder every 15 m in hazardous excavation

✓ Ladder extending 1 m above edge

✓ Safe walkways

✓ Designated entry and exit

Never

✗ Jump into excavation

✗ Climb unstable walls

MODULE 15: EXCAVATION NEAR POWER LINES

मॉड्यूल 15: HT लाइन के पास खुदाई

Rules

- ✓ Maintain electrical clearance
- ✓ Consult electrical department
- ✓ Verify tower stability

Never

- ✗ Operate excavator near HT line without clearance

MODULE 16: EXCAVATION NEAR RAILWAY TRACK

मॉड्यूल 16: रेलवे ट्रैक के पास खुदाई

Requirements

- ✓ Obtain approval
- ✓ Ensure track stability
- ✓ Follow owner department instructions

Never

- ✗ Undermine track foundation

MODULE 17: EXCAVATION AS CONFINED SPACE

मॉड्यूल 17: कन्फाइंड स्पेस जैसी खुदाई

Some excavations may become confined spaces.

Hazards

- ✓ Toxic gas
- ✓ Oxygen deficiency
- ✓ Poor ventilation

Additional Requirements

- ✓ Gas Testing

- ☒ Monitoring
- ☒ Rescue Plan

MODULE 18: EMERGENCY RESCUE

मॉड्यूल 18: आपातकालीन बचाव

Rescue Equipment

- ☒ Breathing Apparatus
- ☒ Safety Harness
- ☒ Lifeline
- ☒ Basket Stretcher
- ☒ First Aid Box

Rule

Emergency rescue equipment must be available before work starts.

MODULE 19: INSPECTIONS

मॉड्यूल 19: निरीक्षण

Inspection Required

- ☒ Daily
- ☒ Before each shift
- ☒ After heavy rain

Check For

- ☒ Cracks
- ☒ Water accumulation
- ☒ Soil condition

- ✓ Shoring condition
- ✓ Barricading
- ✓ Adjacent structures

MODULE 20: PPE REQUIREMENTS

मॉड्यूल 20: PPE आवश्यकताएँ

Mandatory PPE

- ✓ Safety Helmet
- ✓ Safety Shoes
- ✓ Reflective Jacket
- ✓ Gloves
- ✓ Safety Goggles

Additional PPE when required:

- ✓ Dust Mask
- ✓ Respirator
- ✓ Harness

TEN GOLDEN RULES

दस सुनहरे नियम

1. Obtain excavation clearance.
2. Perform trial trenching.
3. Identify underground utilities.
4. Keep spoil piles away from edge.
5. Use barricading.

6. Never enter unsafe excavation.
7. Provide proper access.
8. Inspect after rainfall.
9. Use appropriate PPE.
10. Stop work if conditions become unsafe.